Surrogate Replication of the Correlation-Driven $q = 1/2$ Charge Order in the Kagome Superconductor $CeRu_3Si_2$

(Gerguri *et al.*, arXiv:2603.27408, 2026)

Automated replication (Hermes subagent) — tight-binding + mean-field surrogate

July 20, 2026

Abstract

We test the headline claim of Gerguri *et al.* (2026): that DFT+U reproduces the experimentally *dominant* $q = 1/2$ (Pmma) charge order in $CeRu_3Si_2$ only for a Ce-$4f$ Hubbard interaction $U > 6$ eV, with the weaker $q = 1/3$ (Imma) order remaining nearly degenerate, while treating Ce-$4f$ as core fails to stabilize $q = 1/2$. Full DFT+U is scoped out; instead we build a from-scratch kagome tight-binding + Hartree mean-field *surrogate* (Ru kagome layer hybridized with a heavy Ce-$4f$ flat level) and compare the charge-order susceptibility χ_q at $q = 1/2$, $1/3$, $1/4$ versus U. The surrogate reproduces the paper's mechanism qualitatively: $q = 1/3$ wins at small U, a crossover to $q = 1/2$ occurs at $U \approx 5$ (paper: 6 eV) with the two nearly degenerate near the crossover, and the Ce-$4f$-as-core control yields a $q = 1/4$ (CO*) winner with $q = 1/3$ suppressed — matching the paper's reported failure mode. Verdict: **PARTIAL** (qualitative-only; no first-principles total energies).

1 Paper claim and target

The parent phase is P6/mmm kagome. Synchrotron XRD reveals a dominant $q = 1/2$ charge order (describable in Pmma) plus a weaker $q = 1/3$ component (Imma, CO-II), persisting to 300 K. First-principles DFT shows coexisting Ru d-orbital kagome flat bands and heavy-fermion Ce^{4+} $4f$ flat bands ($4f^0$, above E_F), with multiple imaginary phonon modes in the $q_z = 1/2$ plane. Crucially, with Ce-$4f$ as valence: at $U = 0$ the $q = 1/3$ order is favored (ground state actually a non-CO distortion); the two orders are *nearly degenerate at* $U = 6$ *eV*; and for $U > 6$ eV the $q = 1/2$ order becomes the lowest-energy ground state. With Ce-$4f$ as core, $q = 1/3$ is suppressed and $q = 1/2$ is not stabilized (a $q = 1/4$ CO* appears).

2 Method (surrogate)

We diagonalize a real-space kagome supercell: 3 Ru sites/cell (nearest-neighbor hopping $t = 1$) plus one Ce-$4f$ level/cell (onsite ε_f, Ru–f hybridization $t_f = 0.30$). The Hubbard U is mapped to the $4f$ flat-band position, $\varepsilon_f = \varepsilon_{f,0} + k_U U$ ($\varepsilon_{f,0} = 0.15$, $k_U = 0.10$), encoding that stronger correlations push the $4f^0$ band further above E_F. A commensurate charge order at $q = 1/n$ is imposed as an onsite Ru modulation $\delta \cos(2\pi x/n)$. The competing order is selected by the mean-field CDW susceptibility (Landau curvature)

$$\chi_q = -\frac{E(+\delta) + E(-\delta) - 2E(0)}{\delta^2}, \qquad E = \tfrac{1}{N}\sum_{\text{occ}} \varepsilon_i, \tag{1}$$

evaluated at filling 0.62 (near the Ru kagome M-point van Hove condition). The largest χ_q is the favored order. Geometry and tight-binding conventions are adapted from `loop_current_kagome_kernel.py`; the occupied-energy / density-matrix machinery follows `loop_current_meanfield_kernel.py` (both credited).

3 Results

U (eV)	ε_f/t	$\chi_{1/2}$	$\chi_{1/3}$	$\chi_{1/4}$	ground state
0	0.15	0.389	**0.448**	0.292	$q = 1/3$
2	0.35	0.350	**0.401**	0.225	$q = 1/3$
4	0.55	0.313	**0.324**	0.179	$q = 1/3$
5	0.65	**0.297**	0.278	0.162	$q = 1/2$
6	0.75	**0.284**	0.233	0.149	$q = 1/2$
7	0.85	**0.274**	0.194	0.140	$q = 1/2$
8	0.95	**0.268**	0.165	0.133	$q = 1/2$
9	1.05	**0.265**	0.145	0.129	$q = 1/2$
Ce-4f as core		0.285	0.109	**0.472**	$q = 1/4$ (CO*)

Table 1: Charge-order susceptibility χ_q vs Hubbard U. $q = 1/3$ wins at small U; crossover to $q = 1/2$ at $U \approx 5$; near-degeneracy at $U \approx 4$–5. The f-as-core control gives a $q = 1/4$ CO* winner with $q = 1/3$ strongly suppressed — reproducing the paper's failure mode.

4 Comparison and verdict

- **Reproduced (qualitative):** (i) $q = 1/3$ favored at $U = 0$; (ii) a correlation-driven crossover to a $q = 1/2$ ground state at large U; (iii) near-degeneracy of $q = 1/2$ and $q = 1/3$ around the crossover; (iv) the f-as-core control fails to stabilize $q = 1/2$ and instead favors $q = 1/4$ (CO*) with $q = 1/3$ suppressed — three independent qualitative agreements with the paper.
- **Not reproduced (quantitative):** the surrogate crossover is at $U \approx 5$ (arbitrary t-units mapped to eV), not a first-principles 6 eV; no true total energies; phonon channel omitted; no self-consistency.
- **Verdict:** PARTIAL — gap: no DFT+U total-energy comparison (scoped out); surrogate is qualitative only.

Coverage: 7/10. Agreement: 7/10.

5 Kernel credit

Physics built on the shared TEXTURES-100 kernels `loop_current_kagome_kernel.py` (kagome geometry, primitive vectors, NN Bloch/real-space construction) and `loop_current_meanfield_kernel.py` (occupied-density / band-energy evaluation). See `report/evidence/`.